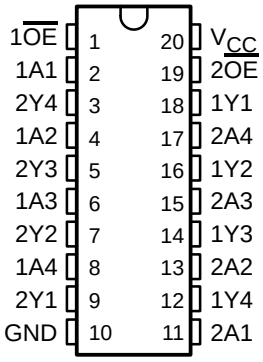
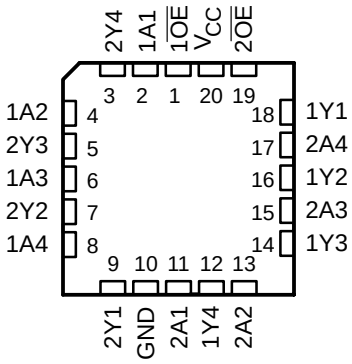
- **Operating Voltage Range of 4.5 V to 5.5 V**
- **State-of-the-Art BiCMOS Design Significantly Reduces $I_{CCZ}$**
- **3-State Outputs Drive Bus Lines or Buffer Memory Address Registers**
- **ESD Protection Exceeds JESD 22**
  - **2000-V Human-Body Model (A114-A)**

**SN54BCT240 . . . J OR W PACKAGE**
**SN74BCT240 . . . DB, DW, N, OR NS PACKAGE**
**(TOP VIEW)**




**SN54BCT240 . . . FK PACKAGE**
**(TOP VIEW)**




## description/ordering information

These octal buffers and line drivers are designed specifically to improve both the performance and density of 3-state memory address drivers, clock drivers, and bus-oriented receivers and transmitters. Together with the ’BCT241 and ’BCT244 devices, these devices provide the choice of selected combinations of inverting and noninverting outputs, symmetrical $\overline{OE}$ (active-low output-enable) inputs, and complementary OE and $\overline{OE}$ inputs. These devices feature high fan-out and improved fan-in.

To ensure the high-impedance state during power up or power down, $\overline{OE}$ should be tied to $V_{CC}$ through a pullup resistor; the minimum value of the resistor is determined by the current-sinking capability of the driver.

**ORDERING INFORMATION**

| $T_A$ | PACKAGE† | | ORDERABLE PART NUMBER | TOP-SIDE MARKING |
|---|---|---|---|---|
| 0°C to 70°C | PDIP – N | Tube | SN74BCT240N | SN74BCT240N |
| | SOIC – DW | Tube | SN74BCT240DW | BCT240 |
| | | Tape and reel | SN74BCT240DWR | |
| | SOP – NS | Tape and reel | SN74BCT240NSR | BCT240 |
| | SSOP – DB | Tape and reel | SN74BCT240DBR | BT240 |
| –55°C to 125°C | CDIP – J | Tube | SNJ54BCT240J | SNJ54BCT240J |
| | CFP – W | Tube | SNJ54BCT240W | SNJ54BCT240W |
| | LCCC – FK | Tube | SNJ54BCT240FK | SNJ54BCT240FK |

† Package drawings, standard packing quantities, thermal data, symbolization, and PCB design guidelines are available at www.ti.com/sc/package.

Please be aware that an important notice concerning availability, standard warranty, and use in critical applications of Texas Instruments semiconductor products and disclaimers thereto appears at the end of this data sheet.

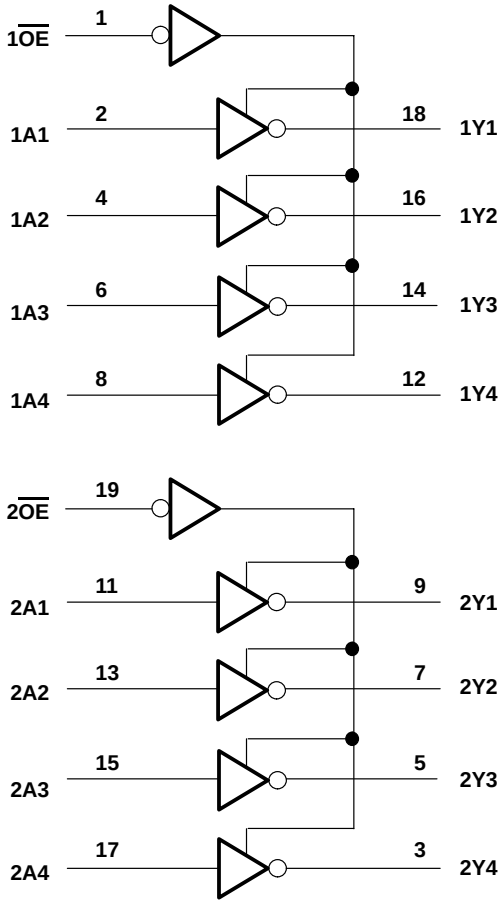
PRODUCTION DATA information is current as of publication date. Products conform to specifications per the terms of Texas Instruments standard warranty. Production processing does not necessarily include testing of all parameters.

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On products compliant to MIL-PRF-38535, all parameters are tested unless otherwise noted. On all other products, production processing does not necessarily include testing of all parameters.

**FUNCTION TABLE**
**(each buffer)**

| INPUTS | | OUTPUT |
|---|---|---|
| $\overline{OE}$ | A | Y |
| L | H | L |
| L | L | H |
| H | X | Z |

## logic diagram (positive logic)



### absolute maximum ratings over operating free-air temperature range (unless otherwise noted)†

Supply voltage range, $V_{CC}$ . . . . . –0.5 V to 7 V
Input voltage range, $V_I$ (see Note 1) . . . . . –0.5 V to 7 V
Voltage range applied to any output in the disabled or power-off state, $V_O$ . . . . . –0.5 V to 5.5 V
Voltage range applied to any output in the high state, $V_O$ . . . . . –0.5 V to $V_{CC}$
Input clamp current, $I_{IK}$ . . . . . –30 mA
Current into any output in the low state: SN54BCT240 . . . . . 96 mA
SN74BCT240 . . . . . 128 mA
Package thermal impedance, $\theta_{JA}$ (see Note 2): DB package . . . . . 70°C/W
DW package . . . . . 58°C/W
N package . . . . . 69°C/W
NS package . . . . . 60°C/W
Storage temperature range, $T_{stg}$ . . . . . –65°C to 150°C

† Stresses beyond those listed under “absolute maximum ratings” may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under “recommended operating conditions” is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

NOTES: 1. The input and output voltage ratings may be exceeded if the input and output current ratings are observed.
2. The package thermal impedance is calculated in accordance with JESD 51-7.

### recommended operating conditions (see Note 3)

| | | SN54BCT240 | | | SN74BCT240 | | | UNIT |
|---|---|---|---|---|---|---|---|---|
| | | MIN | NOM | MAX | MIN | NOM | MAX | |
| $V_{CC}$ | Supply voltage | 4.5 | 5 | 5.5 | 4.5 | 5 | 5.5 | V |
| $V_{IH}$ | High-level input voltage | 2 | | | 2 | | | V |
| $V_{IL}$ | Low-level input voltage | | | 0.8 | | | 0.8 | V |
| $I_{IK}$ | Input clamp current | | | –18 | | | –18 | mA |
| $I_{OH}$ | High-level output current | | | –12 | | | –15 | mA |
| $I_{OL}$ | Low-level output current | | | 48 | | | 64 | mA |
| $T_A$ | Operating free-air temperature | –55 | | 125 | 0 | | 70 | °C |

NOTE 3: All unused inputs of the device must be held at $V_{CC}$ or GND to ensure proper device operation. Refer to the TI application report, *Implications of Slow or Floating CMOS Inputs*, literature number SCBA004.

## electrical characteristics over recommended operating free-air temperature range (unless otherwise noted)

| PARAMETER | TEST CONDITIONS | | SN54BCT240 | | | SN74BCT240 | | | UNIT |
|---|---|---|---|---|---|---|---|---|---|
| | | | MIN | TYP† | MAX | MIN | TYP† | MAX | |
| $V_{IK}$ | $V_{CC}$ = 4.5 V, | $I_I$ = –18 mA | | | –1.2 | | | –1.2 | V |
| $V_{OH}$ | $V_{CC}$ = 4.5 V | $I_{OH}$ = –3 mA | 2.4 | 3.3 | | 2.4 | 3.3 | | V |
| | | $I_{OH}$ = –12 mA | 2 | 3.2 | | | | | |
| | | $I_{OH}$ = –15 mA | | | | 2 | 3.1 | | |
| $V_{OL}$ | $V_{CC}$ = 4.5 V | $I_{OL}$ = 48 mA | | 0.38 | 0.55 | | | | V |
| | | $I_{OL}$ = 64 mA | | | | | 0.42 | 0.55 | |
| $I_I$ | $V_{CC}$ = 5.5 V, | $V_I$ = 7 V | | | 0.1 | | | 0.1 | mA |
| $I_{IH}$ | $V_{CC}$ = 5.5 V, | $V_I$ = 2.7 V | | | 20 | | | 20 | µA |
| $I_{IL}$ | $V_{CC}$ = 5.5 V, | $V_I$ = 0.5 V | | | –1 | | | –1 | mA |
| $I_{OZH}$ | $V_{CC}$ = 5.5 V, | $V_O$ = 2.7 V | | | 50 | | | 50 | µA |
| $I_{OZL}$ | $V_{CC}$ = 5.5 V, | $V_O$ = 0.5 V | | | –50 | | | –50 | µA |
| $I_{OS}$‡ | $V_{CC}$ = 5.5 V, | $V_O$ = 0 | –100 | | –225 | –100 | | –225 | mA |
| $I_{CCH}$ | $V_{CC}$ = 5.5 V, | Outputs open | | 19 | 31 | | 19 | 31 | mA |
| $I_{CCL}$ | $V_{CC}$ = 5.5 V, | Outputs open | | 46 | 71 | | 46 | 71 | mA |
| $I_{CCZ}$ | $V_{CC}$ = 5.5 V, | Outputs open | | 6 | 9 | | 6 | 9 | mA |
| $C_i$ | $V_{CC}$ = 5 V, | $V_I$ = 2.5 V or 0.5 V | | 6 | | | 6 | | pF |
| $C_o$ | $V_{CC}$ = 5 V, | $V_O$ = 2.5 V or 0.5 V | | 11 | | | 11 | | pF |

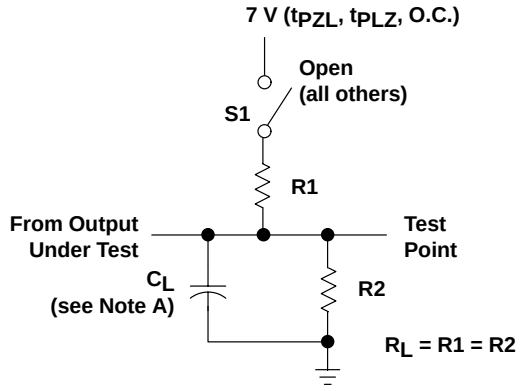
† All typical values are at $V_{CC}$ = 5 V.
‡ Not more than one output should be tested at a time, and the duration of the test should not exceed one second.

## switching characteristics (see Figure 1)

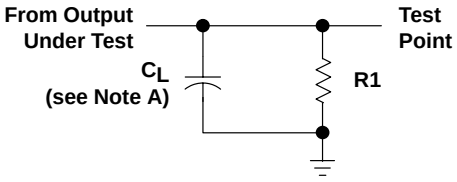
| PARAMETER | FROM (INPUT) | TO (OUTPUT) | $V_{CC}$ = 5 V, $C_L$ = 50 pF, R1 = 500 Ω, R2 = 500 Ω, $T_A$ = 25°C | | | $V_{CC}$ = 4.5 V to 5.5 V, $C_L$ = 50 pF, R1 = 500 Ω, R2 = 500 Ω, $T_A$ = MIN to MAX§ | | | | UNIT |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | 'BCT240 | | | SN54BCT240 | | SN74BCT240 | | |
| | | | MIN | TYP | MAX | MIN | MAX | MIN | MAX | |
| $t_{PLH}$ | A | Y | 0.5 | 3.3 | 4.8 | 0.5 | 6.4 | 0.5 | 5.6 | ns |
| $t_{PHL}$ | | | 0.4 | 1.8 | 3.5 | 0.4 | 4.5 | 0.4 | 4 | |
| $t_{PZH}$ | $\overline{OE}$ | Y | 1 | 6.4 | 7.9 | 1 | 9.2 | 1 | 8.8 | ns |
| $t_{PZL}$ | | | 1 | 7.5 | 9.4 | 1 | 10.8 | 1 | 10.5 | |
| $t_{PHZ}$ | $\overline{OE}$ | Y | 1 | 6 | 6.8 | 1 | 8.5 | 1 | 8.1 | ns |
| $t_{PLZ}$ | | | 1 | 6.7 | 8.1 | 1 | 10.6 | 1 | 9.5 | |

§ For conditions shown as MIN or MAX, use the appropriate value specified under recommended operating conditions.

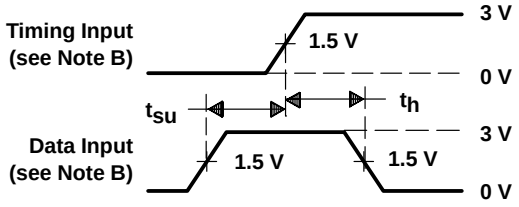
## PARAMETER MEASUREMENT INFORMATION



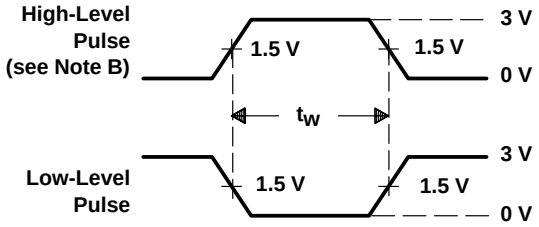

LOAD CIRCUIT FOR
3-STATE AND OPEN-COLLECTOR OUTPUTS



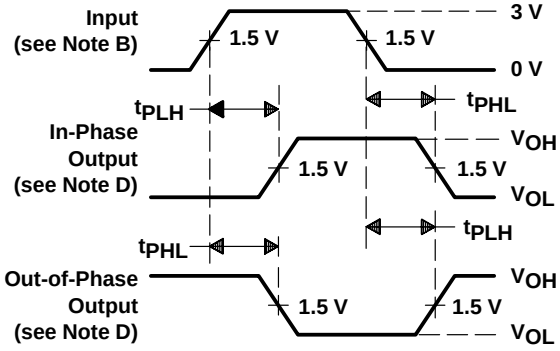

LOAD CIRCUIT FOR
TOTEM-POLE OUTPUTS



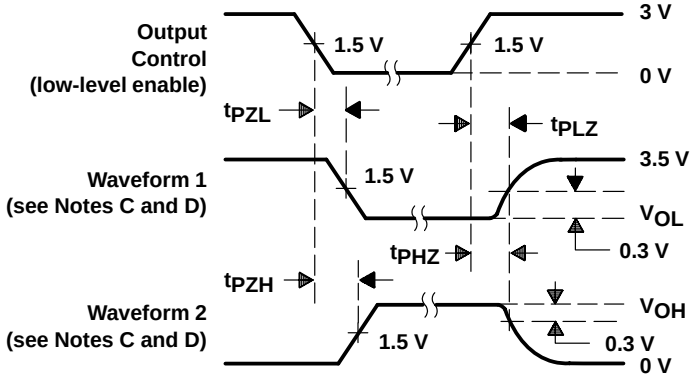

VOLTAGE WAVEFORMS
SETUP AND HOLD TIMES




VOLTAGE WAVEFORMS
PULSE DURATION




VOLTAGE WAVEFORMS
PROPAGATION DELAY TIMES (see Note D)




VOLTAGE WAVEFORMS
ENABLE AND DISABLE TIMES, 3-STATE OUTPUTS

NOTES: A. $C_L$ includes probe and jig capacitance.
B. All input pulses are supplied by generators having the following characteristics: PRR ≤ 10 MHz, $t_r = t_f \leq 2.5$ ns, duty cycle = 50%.
C. Waveform 1 is for an output with internal conditions such that the output is low except when disabled by the output control.
Waveform 2 is for an output with internal conditions such that the output is high except when disabled by the output control.
D. The outputs are measured one at a time with one transition per measurement.
E. When measuring propagation delay times of 3-state outputs, switch S1 is open.
F. All parameters and waveforms are not applicable to all devices.

**Figure 1. Load Circuit and Voltage Waveforms**

## PACKAGING INFORMATION

| Orderable part number | Status (1) | Material type (2) | Package \| Pins | Package qty \| Carrier | RoHS (3) | Lead finish/ Ball material (4) | MSL rating/ Peak reflow (5) | Op temp (°C) | Part marking (6) |
|---|---|---|---|---|---|---|---|---|---|
| 5962-9074201M2A | Active | Production | LCCC (FK) \| 20 | 55 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9074201M2A SNJ54BCT240FK |
| 5962-9074201MRA | Active | Production | CDIP (J) \| 20 | 20 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9074201MRA SNJ54BCT240J |
| 5962-9074201MSA | Active | Production | CFP (W) \| 20 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9074201MSA SNJ54BCT240W |
| SN74BCT240DW | Active | Production | SOIC (DW) \| 20 | 25 \| TUBE | Yes | NIPDAU | Level-1-260C-UNLIM | 0 to 70 | BCT240 |
| SN74BCT240DW.A | Active | Production | SOIC (DW) \| 20 | 25 \| TUBE | Yes | NIPDAU | Level-1-260C-UNLIM | 0 to 70 | BCT240 |
| SN74BCT240N | Active | Production | PDIP (N) \| 20 | 20 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | 0 to 70 | SN74BCT240N |
| SN74BCT240N.A | Active | Production | PDIP (N) \| 20 | 20 \| TUBE | Yes | NIPDAU | N/A for Pkg Type | 0 to 70 | SN74BCT240N |
| SN74BCT240NSR | Active | Production | SOP (NS) \| 20 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | 0 to 70 | BCT240 |
| SN74BCT240NSR.A | Active | Production | SOP (NS) \| 20 | 2000 \| LARGE T&R | Yes | NIPDAU | Level-1-260C-UNLIM | 0 to 70 | BCT240 |
| SNJ54BCT240FK | Active | Production | LCCC (FK) \| 20 | 55 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9074201M2A SNJ54BCT240FK |
| SNJ54BCT240FK.A | Active | Production | LCCC (FK) \| 20 | 55 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9074201M2A SNJ54BCT240FK |
| SNJ54BCT240J | Active | Production | CDIP (J) \| 20 | 20 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9074201MRA SNJ54BCT240J |
| SNJ54BCT240J.A | Active | Production | CDIP (J) \| 20 | 20 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9074201MRA SNJ54BCT240J |
| SNJ54BCT240W | Active | Production | CFP (W) \| 20 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9074201MSA SNJ54BCT240W |

| Orderable part number | Status (1) | Material type (2) | Package \| Pins | Package qty \| Carrier | RoHS (3) | Lead finish/ Ball material (4) | MSL rating/ Peak reflow (5) | Op temp (°C) | Part marking (6) |
|---|---|---|---|---|---|---|---|---|---|
| SNJ54BCT240W.A | Active | Production | CFP (W) \| 20 | 25 \| TUBE | No | SNPB | N/A for Pkg Type | -55 to 125 | 5962-9074201MSA SNJ54BCT240W |

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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**OTHER QUALIFIED VERSIONS OF SN54BCT240, SN74BCT240 :**

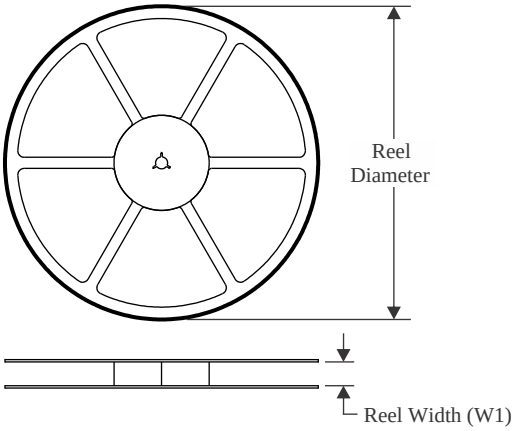
- Catalog : SN74BCT240
- Military : SN54BCT240

NOTE: Qualified Version Definitions:

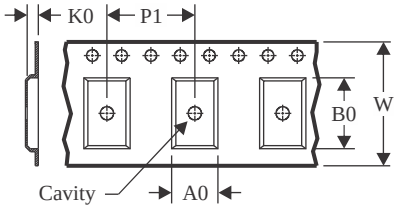
- Catalog - TI's standard catalog product
- Military - QML certified for Military and Defense Applications

## TAPE AND REEL INFORMATION

**REEL DIMENSIONS**

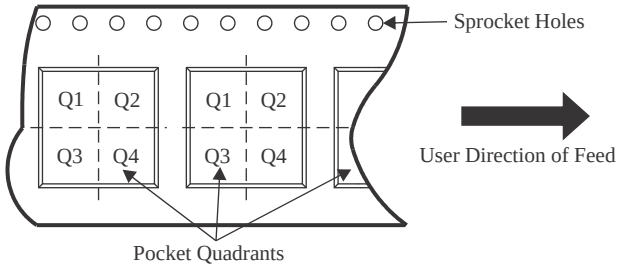



**TAPE DIMENSIONS**




| | |
|---|---|
| A0 | Dimension designed to accommodate the component width |
| B0 | Dimension designed to accommodate the component length |
| K0 | Dimension designed to accommodate the component thickness |
| W | Overall width of the carrier tape |
| P1 | Pitch between successive cavity centers |

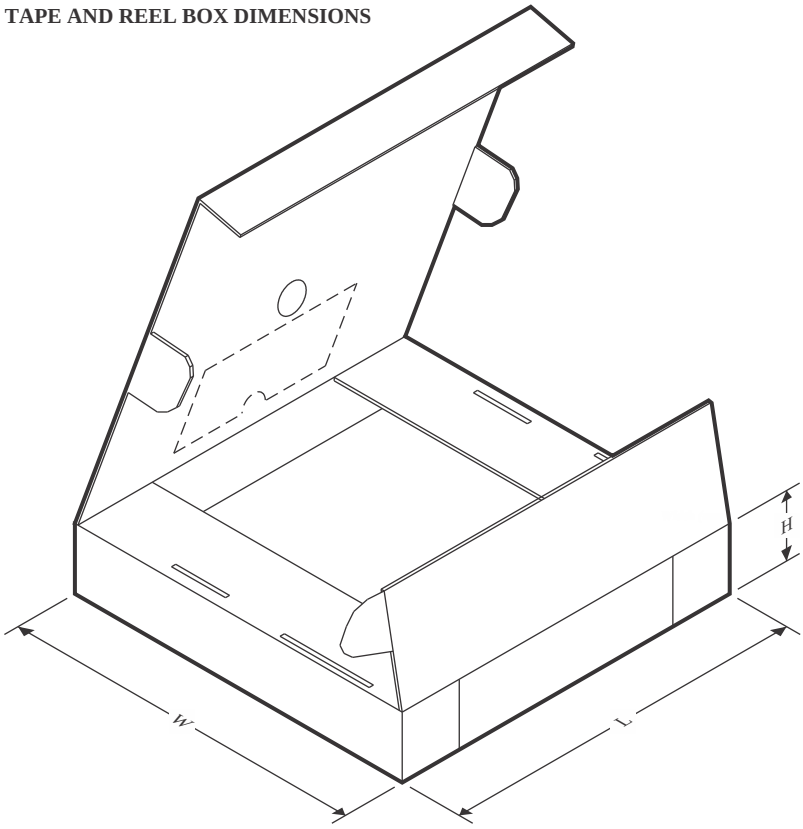
**QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE**




*All dimensions are nominal

| Device | Package Type | Package Drawing | Pins | SPQ | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SN74BCT240NSR | SOP | NS | 20 | 2000 | 330.0 | 24.4 | 8.4 | 13.0 | 2.5 | 12.0 | 24.0 | Q1 |

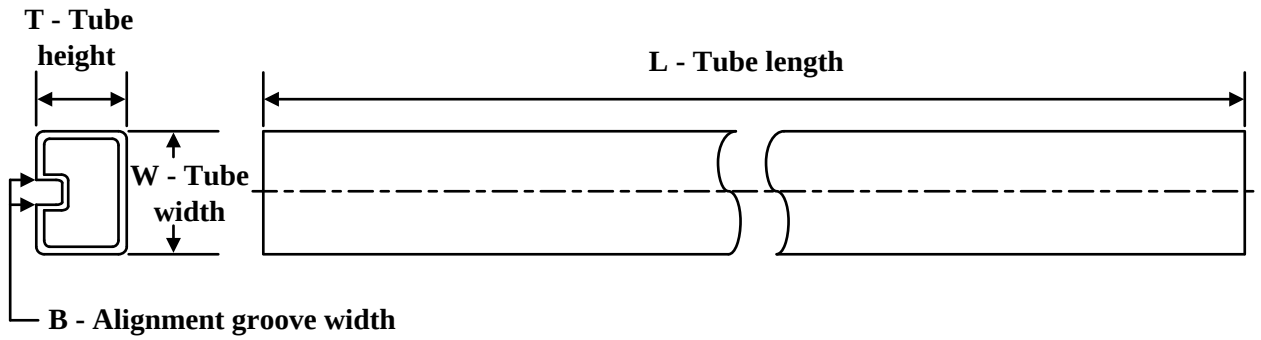
TAPE AND REEL BOX DIMENSIONS




*All dimensions are nominal

| Device | Package Type | Package Drawing | Pins | SPQ | Length (mm) | Width (mm) | Height (mm) |
|---|---|---|---|---|---|---|---|
| SN74BCT240NSR | SOP | NS | 20 | 2000 | 356.0 | 356.0 | 45.0 |

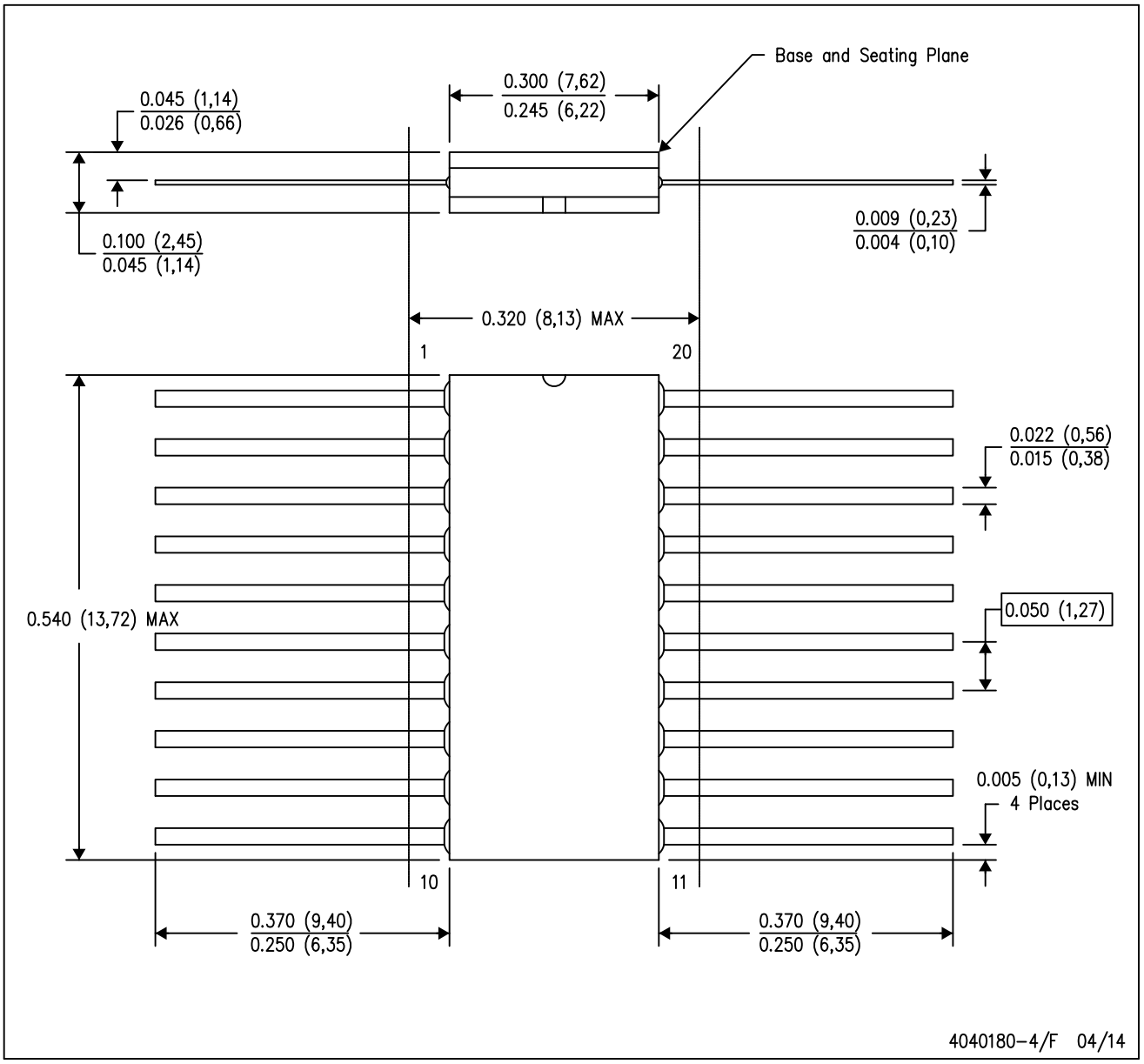
## TUBE




*All dimensions are nominal

| Device | Package Name | Package Type | Pins | SPQ | L (mm) | W (mm) | T (µm) | B (mm) |
|---|---|---|---|---|---|---|---|---|
| 5962-9074201M2A | FK | LCCC | 20 | 55 | 506.98 | 12.06 | 2030 | NA |
| 5962-9074201MSA | W | CFP | 20 | 25 | 506.98 | 26.16 | 6220 | NA |
| SN74BCT240DW | DW | SOIC | 20 | 25 | 507 | 12.83 | 5080 | 6.6 |
| SN74BCT240DW.A | DW | SOIC | 20 | 25 | 507 | 12.83 | 5080 | 6.6 |
| SN74BCT240N | N | PDIP | 20 | 20 | 506 | 13.97 | 11230 | 4.32 |
| SN74BCT240N.A | N | PDIP | 20 | 20 | 506 | 13.97 | 11230 | 4.32 |
| SNJ54BCT240FK | FK | LCCC | 20 | 55 | 506.98 | 12.06 | 2030 | NA |
| SNJ54BCT240FK.A | FK | LCCC | 20 | 55 | 506.98 | 12.06 | 2030 | NA |
| SNJ54BCT240W | W | CFP | 20 | 25 | 506.98 | 26.16 | 6220 | NA |
| SNJ54BCT240W.A | W | CFP | 20 | 25 | 506.98 | 26.16 | 6220 | NA |

W (R–GDFP–F20) CERAMIC DUAL FLATPACK




NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. This package can be hermetically sealed with a ceramic lid using glass frit.
D. Index point is provided on cap for terminal identification only.
E. Falls within Mil–Std 1835 GDFP2–F20

# GENERIC PACKAGE VIEW

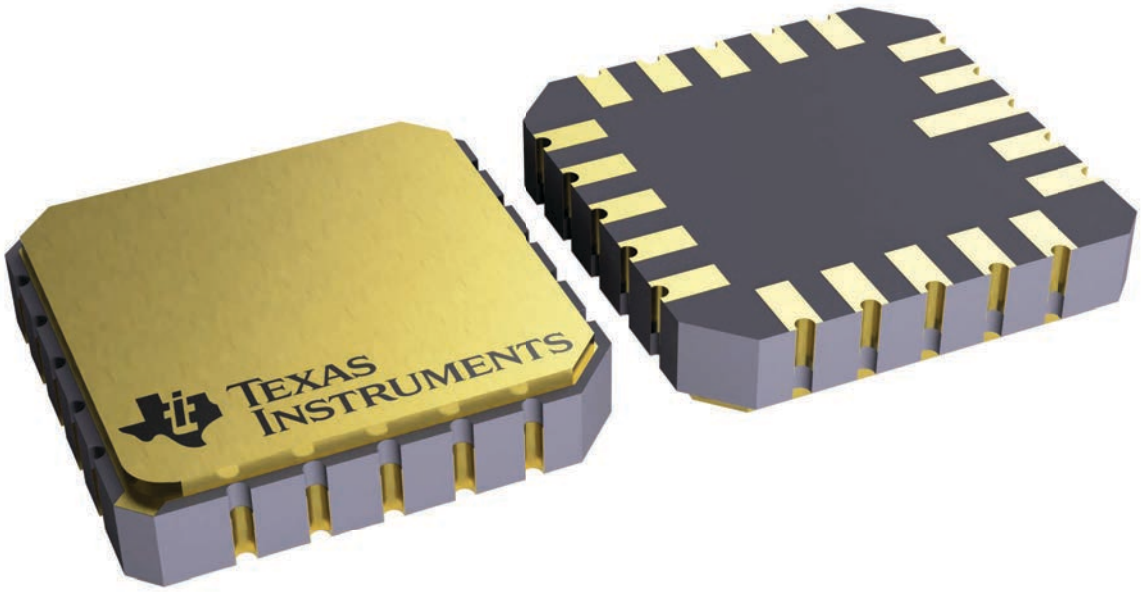
## FK 20

## LCCC - 2.03 mm max height

**8.89 x 8.89, 1.27 mm pitch**

LEADLESS CERAMIC CHIP CARRIER

This image is a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.

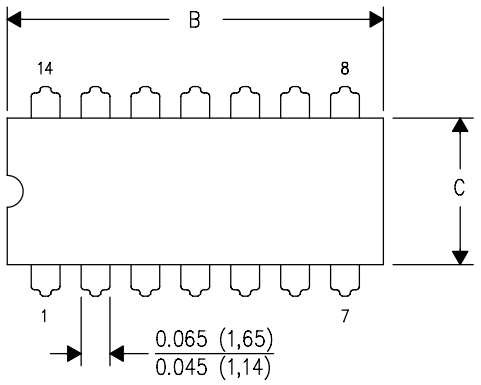



4229370VA\

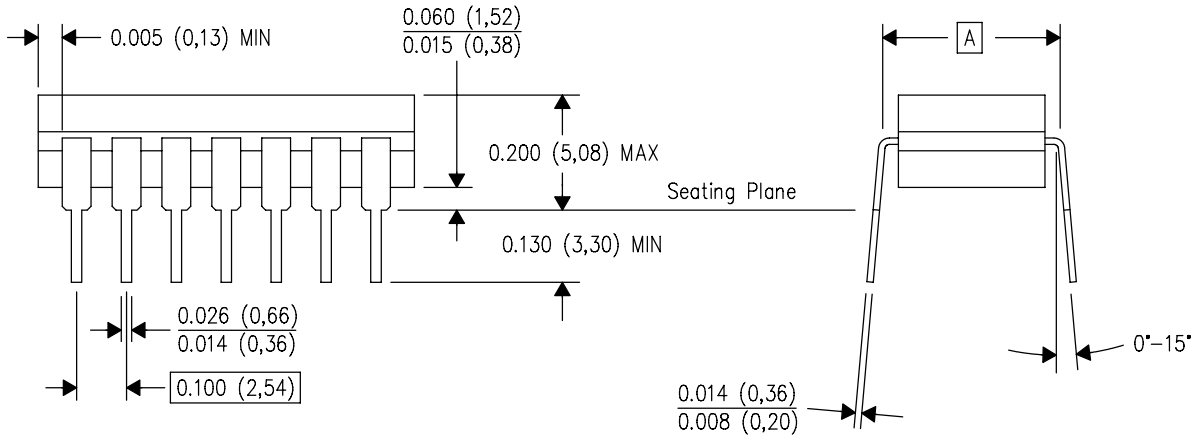
J (R–GDIP–T**)

14 LEADS SHOWN

# CERAMIC DUAL IN–LINE PACKAGE




| DIM \ PINS ** | 14 | 16 | 18 | 20 |
|---|---|---|---|---|
| A | 0.300 (7,62) BSC | 0.300 (7,62) BSC | 0.300 (7,62) BSC | 0.300 (7,62) BSC |
| B MAX | 0.785 (19,94) | .840 (21,34) | 0.960 (24,38) | 1.060 (26,92) |
| B MIN | — | — | — | — |
| C MAX | 0.300 (7,62) | 0.300 (7,62) | 0.310 (7,87) | 0.300 (7,62) |
| C MIN | 0.245 (6,22) | 0.245 (6,22) | 0.220 (5,59) | 0.245 (6,22) |



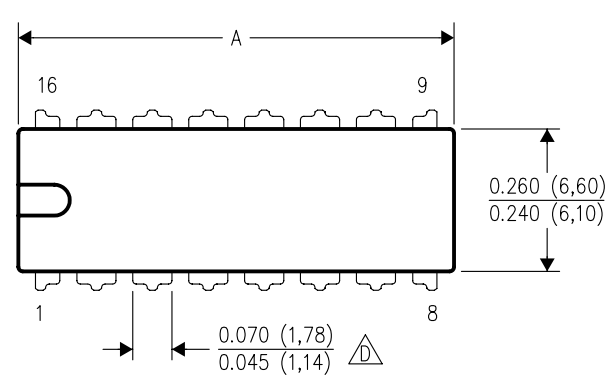

4040083/F 03/03

NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. This package is hermetically sealed with a ceramic lid using glass frit.
D. Index point is provided on cap for terminal identification only on press ceramic glass frit seal only.
E. Falls within MIL STD 1835 GDIP1–T14, GDIP1–T16, GDIP1–T18 and GDIP1–T20.

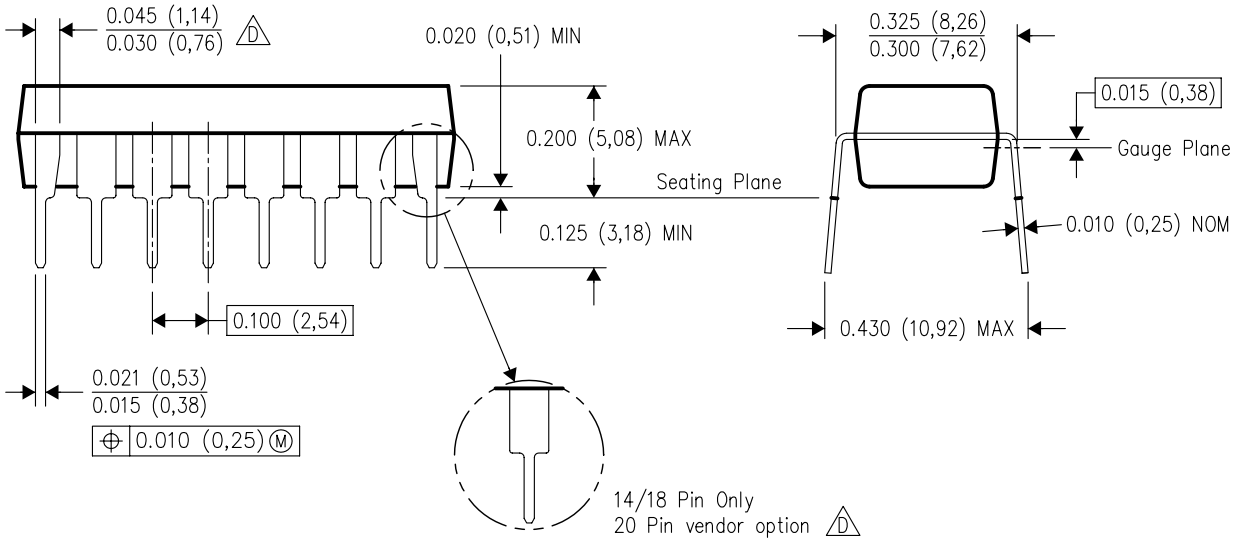
## N (R–PDIP–T**)

## PLASTIC DUAL–IN–LINE PACKAGE

16 PINS SHOWN

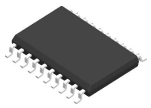



| DIM \ PINS ** | 14 | 16 | 18 | 20 |
|---|---|---|---|---|
| A MAX | 0.775 (19,69) | 0.775 (19,69) | 0.920 (23,37) | 1.060 (26,92) |
| A MIN | 0.745 (18,92) | 0.745 (18,92) | 0.850 (21,59) | 0.940 (23,88) |
| MS–001 VARIATION (C) | AA | BB | AC | AD |




4040049/E 12/2002

NOTES:
A. All linear dimensions are in inches (millimeters).
B. This drawing is subject to change without notice.
C. Falls within JEDEC MS–001, except 18 and 20 pin minimum body length (Dim A).
D. The 20 pin end lead shoulder width is a vendor option, either half or full width.

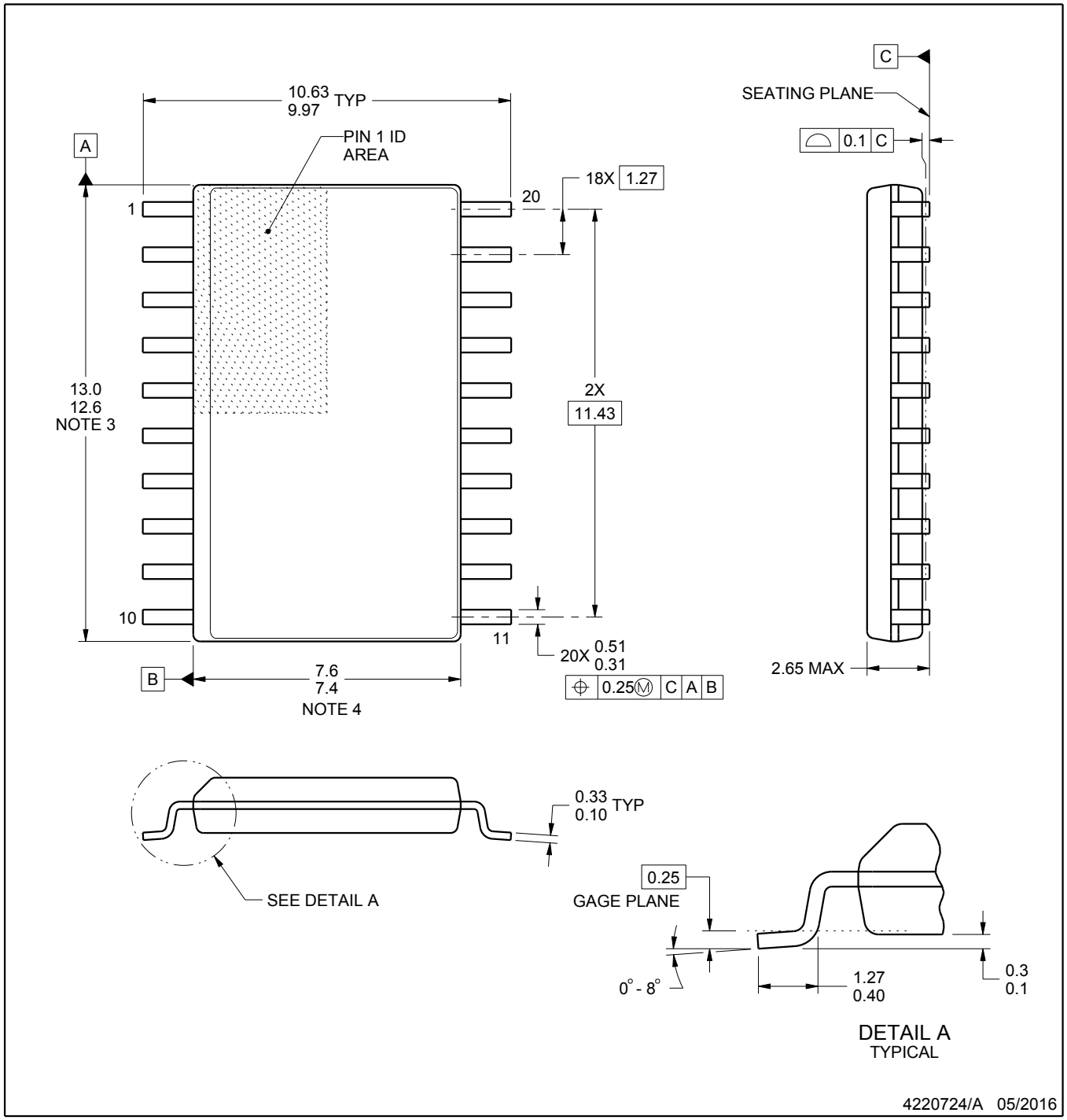


# PACKAGE OUTLINE

## DW0020A

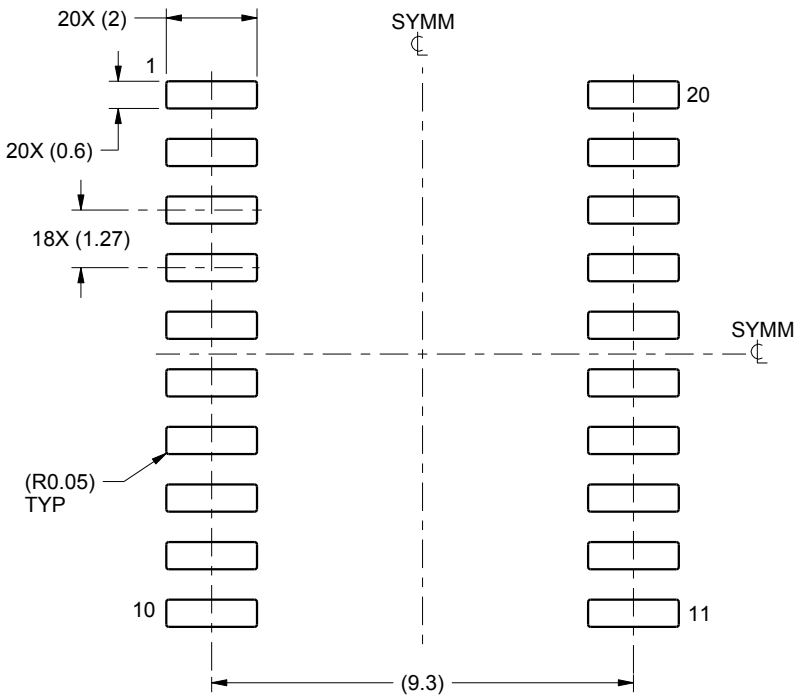
## SOIC - 2.65 mm max height

SOIC

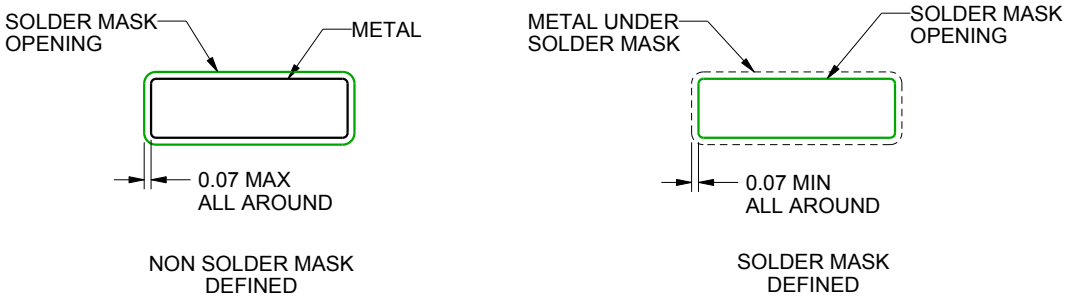



NOTES:

1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.43 mm per side.
5. Reference JEDEC registration MS-013.




LAND PATTERN EXAMPLE
SCALE:6X

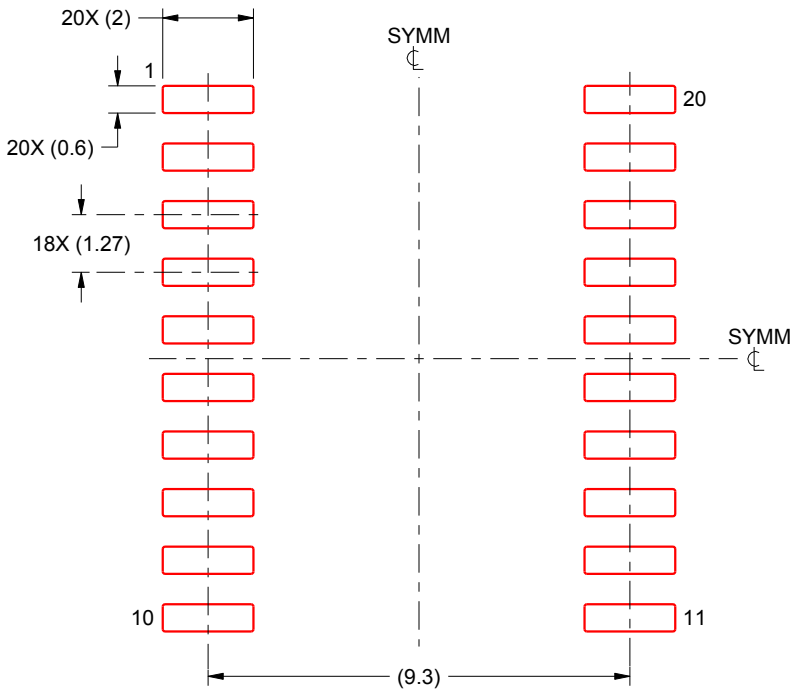



SOLDER MASK DETAILS

4220724/A 05/2016

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.




SOLDER PASTE EXAMPLE
BASED ON 0.125 mm THICK STENCIL
SCALE:6X

4220724/A 05/2016

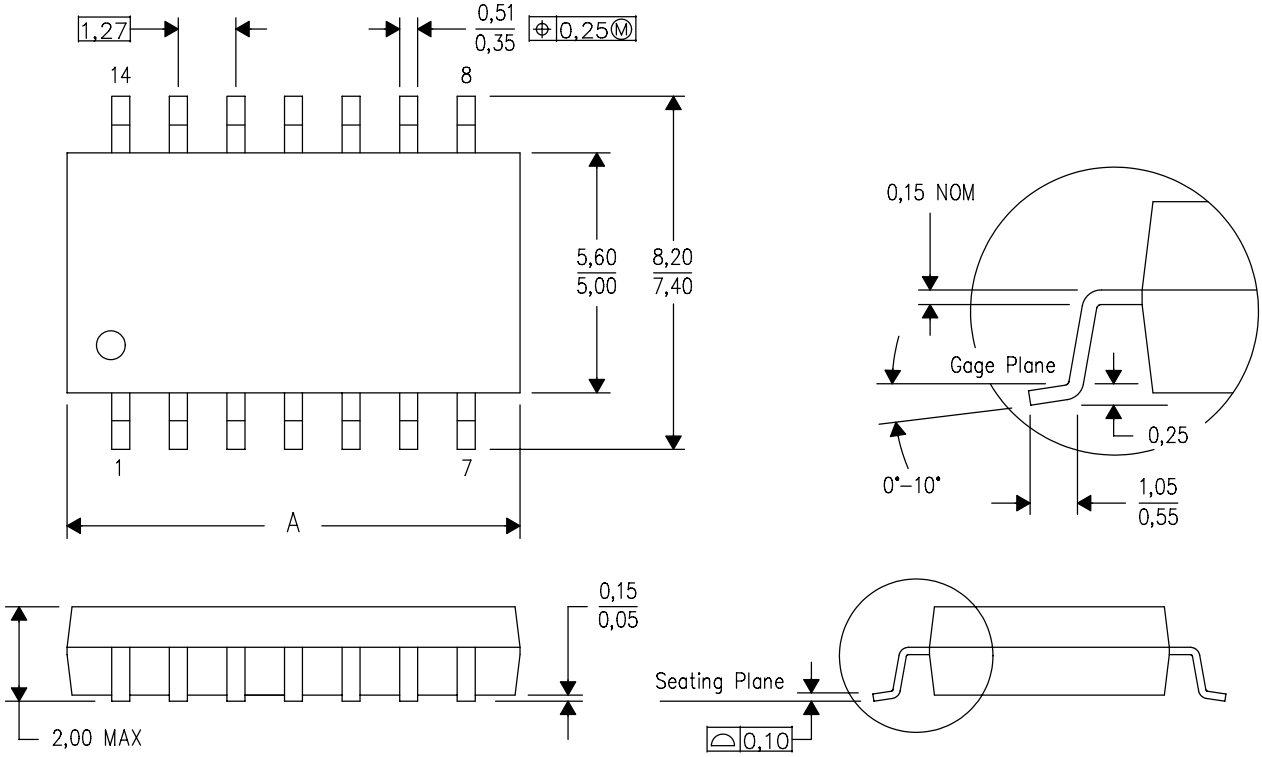
NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

## MECHANICAL DATA

**NS (R-PDSO-G**)** **PLASTIC SMALL-OUTLINE PACKAGE**

**14-PINS SHOWN**




| DIM \ PINS ** | 14 | 16 | 20 | 24 |
|---|---|---|---|---|
| A MAX | 10,50 | 10,50 | 12,90 | 15,30 |
| A MIN | 9,90 | 9,90 | 12,30 | 14,70 |

4040062/C 03/03

NOTES:
A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. Body dimensions do not include mold flash or protrusion, not to exceed 0,15.

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